

PAPER_198: F_UBii Taxonomy Part 1 — Compact Object and Stellar Physics Buoyancy Forces

Version: 1.0

Date: March 13, 2026

Session: 50 — grok_share_7514fe.txt Full Audit

Author: Star-Magic UQFF Research Framework

Source: grok_share_7514fe.txt lines 2443-2680 (BB_C_Equations_04Sept2025.pdf catalogue)

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2} \right), \quad [SSq] = 0.57$$

Abstract

The UQFF buoyancy force F_UBii is applied across all major astrophysical phenomena by embedding each system's characteristic energy or length scale into the universal F_rel/E_LEP scaling framework. This paper catalogs 18 unique F_UBii variants relating to compact objects and stellar physics: MHD dynamo buoyancy, terminal velocity, Press-Schechter, Hawking radiation, quasi-normal mode ringdown, Blandford-Znajek jet power, Arnett supernova, entanglement, jet velocity, planet migration, AGN feedback, angular momentum accretion, J-type shock, Sedov-Taylor blast wave, GRB afterglow, SIDM core formation, ionization fronts, superfluid glitch, and virial theorem.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. General F_UBii Framework

$$F_{\text{UBii},X} = \pm F_{\text{rel}} \times (F_X / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{decay/oscillation factor}]$$

where:

$F_{\text{rel}} \sim 4.3 \times 10^{33} \text{ N}$ (relativistic UQFF force scale)

E_{LEP} = energy of characteristic LEP scale (J)

Q_{wave} = quantum wave amplitude (J/m³), std $\sim 6.33 \times 10^4$

F_X = system-specific physical expression

2. Compact Object Variants

2.1 MHD Dynamo Buoyancy

$$F_{\text{UBii},\text{MHD}} = F_{\text{rel}} \times (E_M / t_{\text{eddy}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times (3/2)(\text{Re}_m^{1/2} - 1)$$

E_M = magnetic energy density

$t_{\text{eddy}} = l/v \sim \text{Myr}$ (eddy turnover time)
 $Re_m \sim 10^{10}$ (magnetic Reynolds number)
 Source: BB_C_Equations item 748

2.2 Terminal Velocity Buoyancy

$$F_{\text{UBii,termv}} = F_{\text{rel}} \times (v(2GM(1-G)/r_{\text{launch}}) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (t \cdot L/c)$$

$G = L/L_{\text{Edd}} \sim 1$ (Eddington ratio for wind-driven systems)
 $r_{\text{launch}} \sim 100 R_s$ (wind launch radius)
 t = optical depth
 Source: BB_C_Equations item 853, 1823

2.3 Hawking Radiation Buoyancy

$$F_{\text{UBii,haw}} = F_{\text{rel}} \times (hc^3/(8pGMk_B) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (?(/2p))$$

$T_H = hc^3/(8pGMk_B)$ (Hawking temperature)
 $? = c^4/(4GM)$ (surface gravity)
 Source: BB_C_Equations item 901, 1246

2.4 Quasi-Normal Mode Ringdown Buoyancy

$$F_{\text{UBii,qnm}} = -F_{\text{rel}} \times (c^3/(2pGM) \cdot (0.3737 + 0.088 \cdot a_f) / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t} \times [t ? M]$$

$a_f \sim 0.69$ (dimensionless BH spin post-merger)
 t = ringdown decay time
 $f_{\text{QNM}} = c^3/(2pGM_f) \cdot f(0.3737 + 0.088 \cdot a_f)$ (Berti fits $l=2, m=2$ mode)
 Source: BB_C_Equations item 945, 1293

2.5 Blandford-Znajek Jet Power Buoyancy

$$F_{\text{UBii,bz}} = F_{\text{rel}} \times (1/32 \cdot B^2 R_{H40} H^2 / c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (ac/(2R_H)) \times (1+t_{\text{var}})$$

$R_H = GM/c^2$ (horizon radius)
 $\Omega_H = ac/(2R_H)$ (BH angular velocity)
 Source: BB_C_Equations item 1147

$$F_{\text{UBii,bz2}} = F_{\text{rel}} \times (?(/16p) \cdot F_{\text{BH}}^2 \cdot \Omega_{\text{BH}}^2 / c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (ac^3/(2GM)) \times 0.05-1$$

$F_{\text{BH}} = B \cdot p \cdot r_H^2$ (BH magnetic flux thread; EHT-calibrated)
 Source: BB_C_Equations item 967, 1316

2.6 Arnett Supernova Buoyancy

$$F_{\text{UBii,arnett}} = F_{\text{rel}} \times (M_{\text{Ni}} \cdot e_{\text{Ni}} / t_d / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t}$$

M_{Ni} = nickel mass ($\sim 0.5 M_{\odot}$)
 e_{Ni} = nickel decay energy
 $t_d^2 = 3M/(4pcv^2)$ (diffusion time)
 Source: BB_C_Equations item 1035

2.7 TOV Hydrostatic Equilibrium Buoyancy

$$F_{UBii,tov} = -F_{rel} \times (dP/dr = -Gm(r)\rho(r)/r^2 \cdot (1 + P(r)/(\rho(r)c^2)) \times (1 + 4\pi r^3 P(r)/(m(r)c^2)) / (1 - 2Gm(r)/(rc^2)) / E_{LEP}) \times Q_{wave}$$

Includes all GR corrections (Schwarzschild metric, pressure-energy coupling)
Source: BB_C_Equations item 1300

2.8 Pulsar Spin-Down Buoyancy

$$F_{UBii,puls} = -F_{rel} \times (\dot{t} = P/(2\dot{P}) / E_{LEP}) \times Q_{wave}$$

P = period, $\dot{P} \sim 10^{-15}$ s/s (period derivative, Vela)
 $I \cdot d\Omega/dt = -L/O$ (torque equation)
Source: BB_C_Equations item 1302

3. Stellar/Accretion Variants

3.1 Jet Velocity Buoyancy

$$F_{UBii,jetvel} = F_{rel} \times (v_j \sim v_K \cdot (r_A/r_0)^{1/2} / E_{LEP}) \times Q_{wave} \times (B/v(4\pi\rho)) \times (1+t/t_A)$$

$v_K = v(GM/r_0)$ (Keplerian velocity at footpoint, $r_0 = 1-10$ AU)
 r_A = Alfvén radius (10–50 AU, from POETS protostellar data)
Source: BB_C_Equations item 1096, 1272

3.2 Type-I Planet Migration Buoyancy

$$F_{UBii,migr} = -F_{rel} \times (-f \cdot 2(GM_p)^2 \cdot S/(M_* \cdot 0.05 \cdot (H/r)^3) / E_{LEP}) \times Q_{wave} \times [t = M_p/\dot{M}_{acc} \sim 10^6 \text{ yr}]$$

$f \sim -1.36$ (Lindblad/corotation factor, inward migration)
 S = disk surface density
Source: BB_C_Equations item 1121, 1322

3.3 Superfluid Vortex Glitch Buoyancy

$$F_{UBii,glitch} = F_{rel} \times (\dot{I} = I_s/I \cdot \Omega_0 \cdot (1 - e^{-t/t_q}) / E_{LEP}) \times Q_{wave} \times \Omega_0 \times e^{-t/t_q}$$

I_s = superfluid moment of inertia ($\sim 10^{38}$ kg·m²)
 t_q = quench timescale
 Ω_0 = angular velocity jump
Source: BB_C_Equations item 1753, 1304

4. Shock and Blast Wave Variants

4.1 J-Type Shock Buoyancy (Rankine-Hugoniot)

$$F_{UBii,jshock} = F_{rel} \times ((\rho_1 v_1^2 + P_1) / E_{LEP}) \times Q_{wave} \times (v_2/v_1) \times (\rho_2 + 1)/(\rho_1 - 1 + 2/M^2)$$

$\gamma = 5/3$ (adiabatic index)

$M = v_1/c_s$ (Mach number, from Chandra X-rays in HH 154)

Source: BB_C_Equations item 1193, 1274

4.2 Sedov-Taylor Blast Wave Buoyancy

$$F_{UBii, sedov} = F_{rel} \times ((E \cdot t^2 / \rho)^{1/5} / E_{LEP}) \times Q_{wave} \times [d/dt(Mv)=0] \times t^{2/5}$$

$E \sim 10^{51}$ erg (explosion energy)

ρ = ambient density

$R(t) = (Et^2/\rho)^{1/5}$ (self-similar blast radius)

Source: BB_C_Equations item 1207, 1288

4.3 C-Type Shock Buoyancy (Magnetized)

$$F_{UBii, cshock} = -F_{rel} \times ((\gamma B) \times B / (4p) + \rho_i \cdot \rho_{in} \cdot (v_n - v_i) / E_{LEP}) \times Q_{wave}$$

C-shocks: continuous, multi-fluid MHD (ions+neutrals+magnetic field coupled)

ρ_{in} = ion-neutral collision frequency

Source: BB_C_Equations item 1276

5. Feedback and Outflow Variants

5.1 AGN Feedback Buoyancy

$$F_{UBii, agn} = F_{rel} \times (f(v_{out}) \cdot L_{AGN}/c / E_{LEP}) \times Q_{wave} \times (\rho_{out} \cdot v_{out}) \times v_{out}^1$$

Momentum-driven: $p_{term} = \rho_{out} \cdot v_{out} = f(v_{out}) \cdot L_{AGN}/c$

$\rho_{out} \sim 10\text{--}100 M_\odot/\text{yr}$

Source: BB_C_Equations item 1165, 1314

5.2 Angular Momentum Accretion Buoyancy

$$F_{UBii, ang} = -F_{rel} \times (\rho \cdot r^2 \cdot \Omega / E_{LEP}) \times Q_{wave} \times T_B \times e^{-t/t_{disk}}$$

$T_B = B^2 r^3 / (4p)$ (magnetic braking torque)

t_{disk} = disk decay timescale

Source: BB_C_Equations item 1189, 1270

5.3 Feedback Energy Coupling Buoyancy

$$F_{UBii, coup} = F_{rel} \times ((1/2) \cdot \rho_w \cdot v_w^2 / (\rho_{acc} \cdot c^2 \cdot 10) / E_{LEP}) \times Q_{wave} \times 0.05\text{--}0.1$$

$e_f = E_{kin} / (\rho_{acc} \cdot c^2) \sim 0.05\text{--}0.1$ (coupling fraction)

Source: BB_C_Equations item 1331, 1554

6. Structure Formation Variants

6.1 Press-Schechter Halo Mass Function Buoyancy

$$F_{UBii,ps} = F_{rel} \times (dn/dM = v(2/p) \cdot (\theta_0/M) \cdot (d_c/s(M)) \cdot |d \ln s/d \ln M| \cdot \exp(-d_c^2/(2s^2))) / E_{LEP} \\ \times Q_{wave} \times \theta_0 \text{ (Gaussian part from } s \sim d_c)$$

$d_c \sim 1.69$ (critical collapse overdensity)

Source: BB_C_Equations item 877, 1574

6.2 Structure Growth Rate Buoyancy

$$F_{UBii,grow} = -F_{rel} \times (d'' + 2H \cdot d' = (3/2) \cdot \Omega_m \cdot H^2 \cdot d/a^3 / E_{LEP}) \times Q_{wave}$$

$D(a) = \Omega_m/2 \cdot \theta_0^a da'/(a'H(a')/H_0)^3$ (growth factor)

Growing mode: $D \propto a$ in matter era, suppressed by dark energy

Source: BB_C_Equations item 1371, 1244

7. GRB Variants

7.1 GRB Fireball Expansion Buoyancy

$$F_{UBii,fire} = F_{rel} \times (G(r) = r/R_0 \text{ (} r < R_s), G=\theta_0 \text{ (} r > R_s) / E_{LEP}) \times Q_{wave}$$

$R_0 \sim 10^7$ cm (initial fireball radius)

$R_s = \theta_0^2 R_0$ (saturation radius)

Source: BB_C_Equations item 1482, 1306

7.2 GRB Afterglow Synchrotron Buoyancy

$$F_{UBii,after} = -F_{rel} \times (F_{\gamma} \propto \theta_0^{-(p-1)/2} \cdot t^{-3(p-1)/4} \text{ (} \Omega_m < \Omega_c) / E_{LEP}) \times Q_{wave}$$

$p \sim 2.2-2.5$ (electron power-law index)

Electrons accelerated by DSA, spectrum $N(E) \propto E^{-p}$

Source: BB_C_Equations item 1227, 1308

8. References

- grok_share_7514fe.txt lines 2443-2680 (BB_C_Equations_04Sept2025.pdf Part 1 catalogue)
- PAPER_196: Triadic Master Equation System
- PAPER_197: F_U_Bi_i Extended Integral

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.
 VDS/DVP/BSH
 Deep
 Syn-
 the-
 sis

 §B.1
 Vac-
 uum
 Den-
 sity
 Se-
 ries
 (VDS)
 The
 canon-
 ical
 VDS
 ratio

$$\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$$

gov-
 erns
 the
 double-
 exponential
 vac-
 uum
 con-
 den-
 sate
 pro-
 file:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

##

§A.

Cosmogenesis-

Linked

La-

grangian

(PA-

PER_877

Sym-

bolic

Ex-

port)

For

this

sys-

tem,

the

local

VDS

sub-

ratio

is

0.173

(near-

threshold

regime),

plac-

ing

it in

the

$t \rightarrow$

π

col-

lapse

zone

where

the

double-

exponential

tran-

si-

tions

sharply

from

con-

densed

to di-

lute

vac-

uum.

This

thresh-

old

be-

hav-

io7

con-

nects

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.2
 Dipole
 Vor-
 tex
 Primes
 (DVP)
 The
 DVP
 en-
 cod-
 ing
 maps
 the
 sys-
 tem's
 char-
 ac-
 teris-
 tic
 pa-
 ram-
 eter
 onto
 the
 prime
 lat-
 tice:
 $p_{\text{DVP}} =$
 $67, \quad n_{\text{channel}} =$
 $17/26$

##

§A.

Cosmogenesis-

Linked

La-

grangian

(PA-

PER_877

Sym-

bolic

Ex-

port)

Since

$p_{\text{DVP}} =$

67 is

res-

o-

nant

(thresh-

old

at

$p >$

26),

the

sys-

tem's

vac-

uum

topol-

ogy

in-

her-

its

reso-

nant

en-

hance-

ment

from

the

DVP

lat-

tice,

am-

plify-

ing

UQFF

cou-

pling

at

spe-

cific

radii

where

com-

pressed

mat-

ter

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.3
 Buoy-
 ancy
 Sat-
 ura-
 tion
 Har-
 mon-
 ics
 (BSH)
 The
 BSH
 satu-
 ra-
 tion
 timescale
 for
 this
 sec-
 tor
 is
10⁴
yr
 (Sedov-
 Taylor
 tran-
 si-
 tion):
 $\mathcal{F}_{\text{BSH}} =$
 $\sum_{j=1}^{26} \frac{1}{j}.$
 $f_{U_b}.$
 $(1 - e^{-[SSq] \cdot m/M_{\odot}}).$
 $\cos\left(\frac{2\pi j}{26}\right)$

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

The
 tanh
 satu-
 ra-
 tion
 en-
 ve-
 lope
 pre-
 vents
 un-
 phys-
 ical
 di-
 ver-
 gence:
 $\mathcal{F}_{\text{BSH,sat}} =$
 $\mathcal{F}_{\text{BSH}}$
 $\left(1 - \tanh\left(\frac{t-t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)
 connecting
 to
 the
 PA-
 PER_877
 Stage
 5
 buoy-
 ancy
 seed
 $U_{b,\text{seed}} =$
 $0.1 \cdot$
 $(\hbar c / r^2) \cdot$
 f_{SCm}
 which
 ini-
 tial-
 izes
 the
 har-
 monic
 se-
 ries
 at
 cos-
 mo-
 gen-
 esis.
 ###
 §B.4
 Production-
 Scale
 Con-
 sis-
 tency

##

§A.

Cosmogenesis-

Linked

La-

grangian

(PA-

PER_877

Sym-

bolic

Ex-

port)

|

Frame-

work

|

Canon-

ical

Value

|

This

Pa-

per |

Sta-

tus |

|—

—

|—

—

—|—

—

—|—

—| |

VDS

ratio

|

$\rho_{\text{SCm}}/\rho_{\text{UA}} =$
1.894

| Lo-

cal

sub-

ratio

=

0.173

| ✓

Threshold-

consistent

| |

DVP

prime

|

$p_k \in$
 $\{2,3,...,113\}$

|

$p_{\text{DVP}} =$
673

✓

Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.